SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

Time Duration: 30 Minutes

QUESTIONS: 30 **Total Marks:30**

Annexure A

MCQ

Code of Conduct:

I ***JEROLIN MARY R (192411024)*** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Circle the most appropriate option for each question.

Q1. Which of the following best describes a Logical Agent in AI?	
a) An agent that reacts randomly to the environment	c) An agent that only follows pre-defined scripts
b) An agent that uses a knowledge base and logical inference to decide actions	d) An agent that learns from rewards

Q2. In Propositional Logic, 'P → Q' is logically equivalent to:	
a) $P \land \neg Q$	c) $\neg Q \rightarrow \neg P$ only
b) $\neg P \lor Q$	d) Both b and c

Q3. Conjunctive Normal Form (CNF) requires the formula to be expressed as:	
a) A disjunction of conjunctions	c) A conjunction of literals only
b) A conjunction of disjunctions (clauses)	d) A sequence of implications

Q4. In FOL, the sentence $\forall x$: Student(x) → Passes(x) means:	
a) Some students pass	c) If something passes, it is a student
b) Every entity that is a student passes	d) There exists at least one student who passes

Q5. Unification in FOL is the process of finding substitutions that:	
a) Convert clauses to CNF	c) Prove a goal using backward chaining
b) Make two logical expressions syntactically identical	d) Derive facts using Modus Ponens

Q6. In the Resolution algorithm, deriving the empty clause □ indicates:	
a) The knowledge base is consistent	c) A contradiction — the original goal is proven
b) The negated goal is satisfiable	d) More clauses need to be added

Q7. Forward Chaining is best described as a ________ strategy.

a) Goal-driven / top-down

b) Data-driven / bottom-up

c) Heuristic-based depth-first

d) Probabilistic inference

Q8. Backward Chaining begins reasoning from:

a) All known facts toward a conclusion

b) The goal and works backward to verify supporting facts

c) A random clause in the knowledge base

d) The resolution refutation tree

Q9. Which step is NOT part of converting a sentence to CNF?

a) Eliminate biconditionals (↔)

b) Move negations inward using De Morgan's laws

c) Apply Skolemization to remove existential quantifiers

d) Distribute AND over OR

Q10. Knowledge Engineering in FOL involves:

a) Training a neural network on domain-specific data

b) Defining domain concepts, relations, and axioms formally in FOL

c) Writing heuristic search strategies

d) Encoding reward functions for reinforcement learning

Q11. Modus Ponens states: If $P \rightarrow Q$ is known and P is true, then:

a) $\neg Q$ is true

b) Q must be true

c) P is uncertain

d) The KB is inconsistent

Q12. The Most General Unifier (MGU) is the substitution that:

a) Is the most specific possible match

c) Eliminates all variables from a clause

b) Makes two terms identical with minimal variable binding

d) Applies resolution to produce CNF

Q13. Which inference method is more appropriate when a specific goal is known?

a) Forward Chaining

b) Truth Table method

c) Backward Chaining

d) A* search

Q14. Propositional Logic differs from First Order Logic because it:

a) Supports probabilistic reasoning

b) Cannot represent objects, relations, or quantifiers — only truth values of atomic propositions

c) Is always more expressive

d) Handles uncertainty better

Q15. Resolution Refutation proves a goal G by:

a) Directly deriving G from the KB using forward chaining

b) Adding ¬G to the KB and deriving a contradiction (empty clause □)

c) Checking G in the truth table

d) Applying Modus Tollens repeatedly

Q16. Learning from observation in AI refers to:

a) An agent receiving numerical rewards from the environment

b) Improving performance by examining examples or experience from the environment

c) Acquiring knowledge by following explicit symbolic rules

d) Memorizing a fixed dataset without generalization

Q17. Inductive learning is best described as:

a) Deriving specific conclusions from general rules

b) Generalizing a hypothesis from specific training examples

c) Memorizing training data exactly

d) Applying search algorithms to find optimal policies

Q18. In a Decision Tree, what does each internal node represent?

a) A class label / output decision
b) A probability distribution over outcomes
c) A test on an attribute / feature
d) A training data sample

Q19. Explanation-Based Learning (EBL) differs from inductive learning because it:

a) Requires large amounts of training data to generalize
b) Learns only from numeric data using regression
c) Uses prior domain knowledge to explain and generalize a single example
d) Applies unsupervised clustering to find hidden patterns

Q20. In statistical learning, a Naive Bayes classifier assumes:

a) All features are conditionally dependent on each other given the class
b) The data must follow a uniform distribution
c) All features are conditionally independent given the class label
d) Labels are determined by a reward function

Q21. Learning with complete data (no hidden variables) in a Bayesian network involves:

a) Using the EM algorithm to estimate missing values
b) Training a deep neural network with dropout regularization
c) Directly computing maximum likelihood estimates from observed data
d) Applying reinforcement signals to update weights

Q22. The Expectation-Maximization (EM) algorithm is used when:

a) Training data is fully labeled and no variables are hidden
b) An agent explores an unknown environment using trial and error
c) The dataset contains hidden (latent) variables that cannot be directly observed
d) Backpropagation fails to converge in a neural network

Q23. In a Neural Network, the activation function serves to:

a) Initialize the weights of each neuron to zero	c) Introduce non-linearity to enable learning of complex patterns
b) Normalize the input dataset before training	d) Store previously learned examples in memory

Q24. Which of the following correctly describes backpropagation in a neural network?

a) Weights are updated in the forward pass using input gradients	c) The error is propagated backward from output to input layers to adjust weights
b) Outputs of each layer are stored and reused in later epochs	d) A separate model is trained for each hidden layer independently

Q25. A Support Vector Machine (SVM) — a type of kernel machine — classifies data by:

a) Minimizing the number of training examples used	c) Finding the maximum-margin hyperplane that separates classes
b) Grouping similar data points into clusters using centroids	d) Updating weights using Q-values from an environment

Q26. The kernel trick in kernel machines (SVMs) allows:

a) Reducing the number of support vectors needed	c) Operating in a high-dimensional feature space without explicitly computing coordinates
b) Accelerating gradient descent convergence	d) Replacing hidden layers in a deep neural network

Q27. In Reinforcement Learning, the agent aims to maximize:

a) Classification accuracy on a fixed test set	c) Cumulative reward received over time through interaction with the environment
b) The number of features selected from the training data	d) The size of the neural network hidden layer

Q28. The Q-Learning algorithm in Reinforcement Learning is classified as:	
a) A supervised learning algorithm requiring labeled datasets	c) A model-free, off-policy temporal difference learning method
b) An unsupervised clustering technique using centroids	d) A decision-tree-based ensemble classifier

Q29. Overfitting in machine learning occurs when:	
a) The model performs poorly on both training and test data	c) The model fits training data too closely and generalizes poorly to new data
b) The learning rate is set too low during gradient descent	d) The training set has more features than training examples

Q30. Which of the following best distinguishes Reinforcement Learning from Supervised Learning?	
a) RL uses labeled input-output pairs; SL uses reward signals	c) RL learns via trial-and-error with reward feedback; SL learns from labeled examples
b) Both RL and SL require explicit reward functions	d) RL is always faster to converge than SL on any dataset

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	30%	Demonstrates thorough understanding; answers all questions accurately	Shows good understanding with few errors	Partial understanding; several incorrect responses	Lacks understanding; majority of answers incorrect
Accuracy of Answers	25%	All answers are correct	Most answers are correct with minor mistakes	Some correct answers; frequent errors	Mostly incorrect answers
Application of Knowledge	20%	Correctly applies concepts to	Applies concepts with minor errors	Limited ability to apply concepts	Unable to apply concepts

		problem-based questions			
Time Management	15%	Completes quiz well within time efficiently	Completes on time with minor delays	Requires extra time or rushes through questions	Unable to complete within given time
Question Interpretation	10%	Interprets all questions correctly and responds appropriately	Misinterprets a few questions	Often misinterprets questions	Unable to understand questions

ANSWER

Q.No	Option	Answer
1	b	An agent that uses a knowledge base and logical inference to decide actions
2	d	Both $\neg P \lor Q$ and $\neg Q \rightarrow \neg P$
3	b	A conjunction of disjunctions (clauses)
4	b	Every entity that is a student passes
5	b	Make two logical expressions syntactically identical
6	c	A contradiction — the original goal is proven
7	b	Data-driven / bottom-up
8	b	The goal and works backward to verify supporting facts
9	c	Apply Skolemization to remove existential quantifiers
10	b	Defining domain concepts, relations, and axioms formally in FOL
11	b	Q must be true
12	b	Makes two terms identical with minimal variable binding
13	c	Backward Chaining
14	b	Cannot represent objects, relations, or quantifiers—only truth values of atomic propositions
15	b	Adding $\neg G$ to the KB and deriving a contradiction (empty clause □)

16	b	Improving performance by examining examples or experience from the environment
17	b	Generalizing a hypothesis from specific training examples
18	c	A test on an attribute / feature
19	c	Uses prior domain knowledge to explain and generalize a single example
20	c	All features are conditionally independent given the class label
21	c	Directly computing maximum likelihood estimates from observed data
22	c	The dataset contains hidden (latent) variables that cannot be directly observed
23	c	Introduce non-linearity to enable learning of complex patterns
24	c	The error is propagated backward from output to input layers to adjust weights
25	c	Finding the maximum-margin hyperplane that separates classes
26	c	Operating in a high-dimensional feature space without explicitly computing coordinates
27	c	Cumulative reward received over time through interaction with the environment
28	c	A model-free, off-policy temporal difference learning method
29	c	The model fits training data too closely and generalizes poorly to new data
30	c	RL learns via trial-and-error with reward feedback; SL learns from labeled examples